

DMPK capabilities: from depletion to exposure

02/18/2025 | R2 | Priya Nandekuri

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Start with the decision

- Which test articles need an exposure explanation before the next study?
- Define species, matrix and intended dose route before selecting the assay sequence.
- A useful DMPK memo connects analytical limits to the exposure gate.

Connected study workflow

Decision question > Intake Characterization > Analytical readiness Exposure > DMPP memo



Intake > In vitro characterization Readiness > Rat exposure

Intake and accession lots

- Capture article identity, accession lot, formulation and handling history.
- Separate material identity from assay readiness.
- Maintain the identifier through assay worksheets and analyte cards.
- Discuss limited material before selecting the study sequence.

Species and matrix plan

- Choose microsomes or hepatocytes according to the clearance question.
- Identify plasma species for $f_{u,p}$ by equilibrium dialysis.
- State the matrix for Rat exposure with tandem-MS bioanalysis.
- A matrix change requires assessment of calibration and recovery assumptions.

Matrix acceptance before ranking

- Set reference conditions and matrix controls in the study plan.
- Review calibration behaviour, blanks and sample suitability together.
- Preserve qualified observations and their reasons.
- Helen Marchwood, Bioanalytical Quality Manager, supports the acceptance framework.

Microsomal CLint

- Measure depletion in a defined microsomal incubation.
- Cofactor dependence and nonspecific loss distinguish metabolism from other loss routes.
- Report CLint in $\mu\text{L}/\text{min}/\text{mg}$ with species and conditions attached.
- Flat or very rapid profiles may support bounds rather than precise estimates.

Hepatocyte depletion CLint

- Use an intact-cell context when the question extends beyond microsomal metabolism.
- Record species, cell handling and the suitable depletion interval.
- Report CLint in $\mu\text{L}/\text{min}/10^6$ cells.
- Do not compare directly with microsomal values without an explicit scaling model.

Choosing the depletion model

- Microsomal CLint: focused assessment of microsomal metabolic loss.
- Compare articles within matched incubation conditions.
- Hepatocyte depletion CLint: integrated intact-cell context.
- Consider when transport or additional metabolic routes affect interpretation.

fu,p by equilibrium dialysis

- Measure unbound fraction in the specified plasma matrix.
- Assess recovery, equilibration and stability alongside binding.
- Low unbound fraction requires suitable analytical sensitivity.
- Carry species and matrix into unbound-exposure calculations.

Tandem-MS bioanalytical readiness

- Confirm analyte response in the intended matrix.
- Choose calibration range around the sampling question.
- Consider dilution, carryover and matrix effects before exposure samples arrive.
- Javier Quintana, Mass-Spectrometry Manager, leads analytical planning.

Lower-limit assumptions

- Define how below-quantitation observations enter summaries before reviewing exposure.
- Distinguish low measured concentrations from qualified observations.
- State imputation or exclusion rules in the DMPK memo.
- A terminal phase dominated by qualified observations may not support extrapolation.

Rat exposure with tandem-MS bioanalysis

- Agree route, formulation and sampling against the exposure question.
- Retain dosing-lot and accession-lot linkage.
- Report integrated exposure in ng·h/mL with the integration convention.
- Megan Larkin, DMPK Study Director, connects analytical interpretation to the decision.

Decision-ready DMPK memo

Record element	Required context
Identity	Article and accession lot
Depletion	Species, model and CLint units
Binding	Plasma matrix and recovery
Bioanalysis	Calibration range and reporting limit
Exposure	Route, sampling and integration assumptions
Decision	Conclusion, qualification and next experiment

Exposure gates: state the assumptions

- Compare articles within matched species, matrix and assay context.
- Use binding and depletion to explain exposure.
- Resolve identity and matrix acceptance questions before named-article recommendations.
- Specify the follow-up experiment when the explanation remains ambiguous.

Planning discussion | Priya Nandekuri

- Bring the test article list, available material and intended exposure decision.
- Agree assay sequence and analytical readiness before study dates.
- Define deliverables: assay worksheets, analyte cards, compound tables and DMPK memo.
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